

Oscar Ndugbu

“Systems you can trust at 2 AM.”

STAFF BACKEND & PLATFORM ENGINEER · AI INFRASTRUCTURE · RELIABILITY ENGINEERING

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◆ SUMMARY

Staff-level backend and platform engineer who designs around explicit failure modes, not happy-path assumptions. Architects multi-tenant fintech infrastructure, ensemble ML serving, and autonomous multi-agent orchestration under real-world constraints. A decade of federal-scale data engineering informs every system: make failure visible, recovery deliberate, decisions inspectable.

◆ CORE COMPETENCIES

LANGUAGES & RUNTIMES

- TypeScript
- Node.js
- Python
- Java 17
- SQL

BACKEND & APIS

- Fastify 5
- FastAPI
- Spring Boot 3
- GraphQL
- gRPC
- BullMQ
- WebSocket

DATA & CLOUD

- PostgreSQL (RLS)
- Redis
- Docker
- Kubernetes
- AWS
- GCP

OBSERVABILITY / SRE

- Prometheus
- Grafana
- OpenTelemetry
- Sentry

ML & AI SYSTEMS

- XGBoost
- LightGBM
- CatBoost
- PSI Drift
- Brier Calibration
- Multi-Agent Orch.
- Ollama
- Claude/GPT-4

SYSTEMS DESIGN

- Multi-Tenant RLS
- RBAC
- Event-Driven
- Circuit Breaker
- AES-256
- ZK-Proofs (Groth16)

◆ PROFESSIONAL EXPERIENCE

Infrastructure Consultant — Backend & Reliability JAN 2024 – PRESENT
Independent Consulting · Remote · retained by fintech & SaaS CTOs

- Built an AI-assisted incident-diagnosis tool that converts server logs and traces into ranked root-cause hypotheses, preserving the full evidence path for engineer review.
- Directed backend performance audits, tracing latency spikes and connection-pool exhaustion to root cause and delivering remediation plans tied to observable bottlenecks.
- Designed proactive monitoring pipelines that surface anomalous behavior with enough context for an engineer to act before a local fault becomes a wider incident.

Lead Backend Engineer — Encrypted Data Platform JAN – JUN 2024
Hashablanca (Contract via Consulting Practice) · Remote · Ethereum, Polygon, BSC, StarkNet

- Engineered the core transfer backend for AES-256-encrypted payloads across four blockchain networks, with retry semantics, checkpointed recovery, and an interruption-safe audit record.
- Integrated Zero-Knowledge (Groth16) proof pipelines to validate payload integrity without exposing underlying data, tested across every integration boundary.

Senior Data & Systems Engineer — Planning, Research & Statistics 2014 – PRESENT
Universal Basic Education Commission (UBEC) · Federal Government · Abuja HQ

- Architected state-isolated ETL pipelines consolidating education statistics from all 36 state offices into one audit-ready database, replacing weeks of manual reconciliation per cycle.
- Designed cross-state validation gates with probabilistic record linkage so 35 completed reports proceed while one delayed submission stays visibly pending.
- Built Python / Power BI dashboards giving senior planners real-time visibility into national education metrics that drive federal budget allocation.

◆ **SYSTEMS ARCHITECTED**

Constraint → Decision → Outcome — four production platforms, designed and operated end to end.

TaxBridge

Multi-Surface Tax Compliance Platform

Next.js 15 · Fastify 5 · Java 17/Spring Boot 3 · PostgreSQL 15 RLS · Redis · BullMQ · GraphQL

CONSTRAINT NRS audit scrutiny demands tenant isolation that holds even when a query path omits its predicate.

DECISION Enforced isolation at the PostgreSQL RLS layer, not application code — plus idempotent BullMQ queues and a hash-chained audit trail.

OUTCOME **4 hrs → 15 min** end-to-end filing for Nigerian SMEs; zero data-loss recovery path.

SabiScore

Ensemble ML Prediction & Observability

Next.js 15 · FastAPI · XGBoost/LightGBM/CatBoost · Redis Pub/Sub · Prometheus · Grafana

CONSTRAINT Concurrent sessions on synchronous REST create a thundering-herd query pattern.

DECISION FastAPI + Redis Pub/Sub with a stale-while-revalidate cache; Brier/PSI drift monitoring triggers retraining; a labeled degraded path replaces stale or opaque failure.

OUTCOME **99.9%+ uptime** (90-day window), **45% MTTD** reduction, sub-150ms p99.

SwarmXQ

Autonomous Multi-Agent Orchestration

Python · FastAPI · Ollama (triadic local routing) · PostgreSQL/pgvector · Redis · BullMQ · WebSocket

CONSTRAINT Self-improving agents must evolve strategy without regressions or duplicated side effects on failure.

DECISION Separated dispatch from evolution — a rewritten strategy is promoted only if it beats the current one on a validation set; checkpointed replay resumes failed sub-tasks.

OUTCOME Self-improving fleet with checkpoint recovery, zero manual tuning, under an **8 GB** inference budget.

The Yap Engine

Structured Short-Form Video Generation

Python 3.11+ · Fastify 5 · Ollama · Redis · BullMQ

CONSTRAINT Long creative-generation jobs on constrained CPU/RAM can silently die into opaque restarts.

DECISION Staged the pipeline with visible stage boundaries, recoverable job state, and explicit queue / timeout / resource-pressure limits.

OUTCOME Publish-ready creative packages with explicit recovery and resource visibility under real hardware limits.

◆ **OPEN SOURCE CONTRIBUTIONS**

pg-tenant **NODE** · **POSTGRES**

Moves tenant isolation to the PostgreSQL RLS boundary so it's independently testable, not scattered across every query path.

audit-chain **FINTECH**

Hash-linked record pattern where any retroactive edit breaks the verification chain at the point of modification.

node-debug-11m **AI** · **DEVOPS**

Ranks root-cause hypotheses from live logs and traces while preserving the full evidence path for engineer review.

◆ **LEADERSHIP & COMMUNITY**

- ▶ Architecture advisor to early-stage fintech founders on system design, ML deployment strategy, and security posture at pre-launch and early-growth stages.
- ▶ Speaker at Lagos JS and Python meetups — distributed systems, production ML deployment, and resilient backend infrastructure.
- ▶ Technical author at scardubu.dev, publishing engineering guides on backend reliability, platform architecture, and AI infrastructure decisions.

◆ **CERTIFICATIONS & EDUCATION**

- ▶ AWS Certified Developer · 2023
- ▶ GCP Associate Cloud Engineer · 2023
- ▶ OpenJS Node.js Services (JSNSD) · 2024
- ▶ PostgreSQL 14 Associate · 2024

B.Tech, Environmental Technology
Federal University of Technology,
Owerri · 2006–2011

◆ **EVIDENCE & RECORD**

- ▶ Public repositories and authored engineering notes: **github.com/Scardubu**
 - ▶ Live product & writing archive: **scardubu.dev**
- Non-CS academic background — certifications verifiable on request; public repositories provide the inspectable engineering record.*